General Assembly

STUDY GUIDE

INTRODUCTION TO JAVASCRIPT

The Basics

- Invented in 1995 by Brendan Eich
- Supported by all mainstream browsers
- Used in conjunction with HTML and CSS
- Fast and flexible because it can respond to user input without communicating with a backend server
- Not the same as Java - it's generally used on the front-end (the part of the web you can see), while Java is mostly used on the back-end

Syntax

Capitalization

- JavaScript is case-sensitive
- Multiple words are written in **camel case** - the first word is lowercase and any subsequent words are uppercase

Punctuation

- Most statements end with a semicolon
- Semicolons used to be required at the end of each statement, but the language has evolved and they're not always necessary

Variables

- Variables hold defined values and function like the nouns of JavaScript
- The parts of a variable:
    - The introductory keyword, which indicates the type of variable that will follow
        - **const** - short for constant, used with a variable that is not going to change
        - **let** - used with variables that will change in value
        - **var** - stands for "variable," used with variables that can be changed. This is how variables used to be defined in JavaScript, but it's not commonly used today. "var" may be seen around in older code.
    - The variable's name, written in camel case, for example, myName
    - The value, which is what the variable represents
- Sample - const myName = "Lisa Simpson";

Functions

- Functions are action-oriented, like verbs - they tell the program what to do with the variables you've defined
- For example, **console.log( )** is a function that sends a message to the console

Data Types

Variables can contain one of the following data types:

- Numbers - JavaScript treats whole numbers and decimals the same way
- Booleans - True or false values
- Strings - Any combination of letters, words, numbers, or characters, contained within quotation marks
    - You can use single quotes or double quotes, but it's best to be consistent throughout your code

Concatenation

Concatenation refers to the act of creating combinations.

- You can combine two variables in one string with a + sign
- If you need anything to come between the two variables, like a space or a string of additional words, be sure to add it in quotation marks.
- Sample - console.log(firstName + " " + lastName);